

DATUM LINE

AI × BIO · 29 SEPT

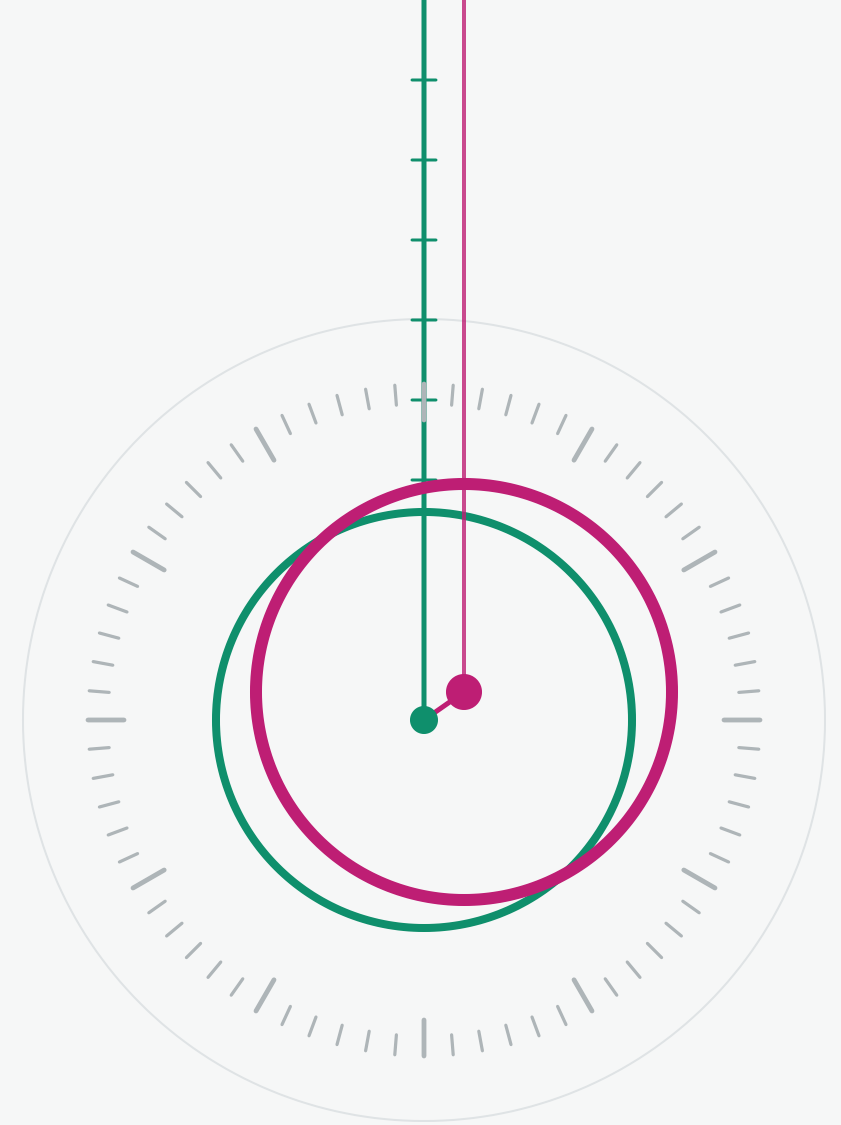
# Control is an illusion.

Confidence without control.

● your cells    ○ the field

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Kat Titterton · San Francisco





[macrofluidic.me/resume](https://macrofluidic.me/resume)

# Katherine Titterton

Five labs, and the same **unanswered question** at every bench. It made the science slow and wrong.

- 2023–2026 **Axiom** SAN FRANCISCO  
founding MTS • hepatocytes at scale
- 2022–2023 **insitro** SOUTH SAN FRANCISCO  
neuroscience assays • productionization
- 2021–2022 **MIT** SYNTHETIC NEUROBIOLOGY  
expansion microscopy
- 2018–2021 **abbvie** NEUROSCIENCE  
tau • neurons
- 2017–2018  HARVARD MEDICAL SCHOOL  
Xenopus oocytes • GABA<sub>A</sub> receptors
- 2017 **COLGATE** BA  
neuroscience + math biology

## MY WORK

- 1 AXIOM BIO  
Ewald JD, **Titterton KL**, et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Systems*, 2026.
- 2 AXIOM BIO  
Seal S, Dee W, ... **Titterton K**, ... Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nature Communications*, 2026.
- 3 ABBVIE  
Wu JW, **Titterton K**, et al. A neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*, 2025.
- 4 MIT  
Kang J, ... **Titterton K**, ... Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nature Communications*, 2024.
- 5 MGH • HARVARD MEDICAL SCHOOL  
Wu B, Jayakar SS, ... **Titterton K**, ... Bruzik KS. Inhibitable photolabeling by a neurosteroid diazine analog in the  $\beta 3$  subunit of GABA<sub>A</sub> receptors. *Eur J Med Chem*, 2019.

## PUBLISHED WITH

**Broad Institute • MIT • AbbVie • EMBL-EBI • HESI • Bayer • University of Cambridge • Uppsala University**

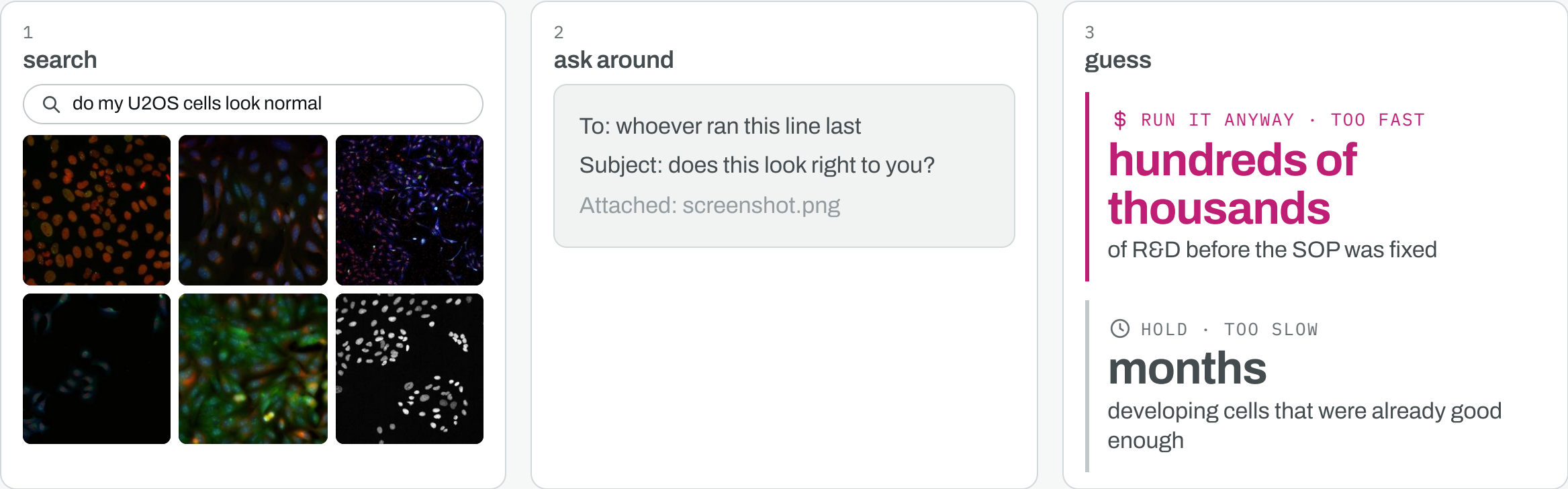
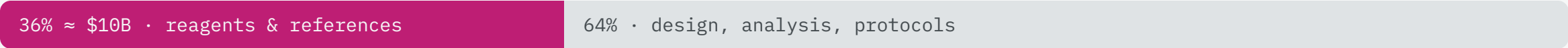
# “Do my cells look normal?”

WHY NOW

Since 2023 the field’s reference is public: JUMP, CC0 [7].

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6]

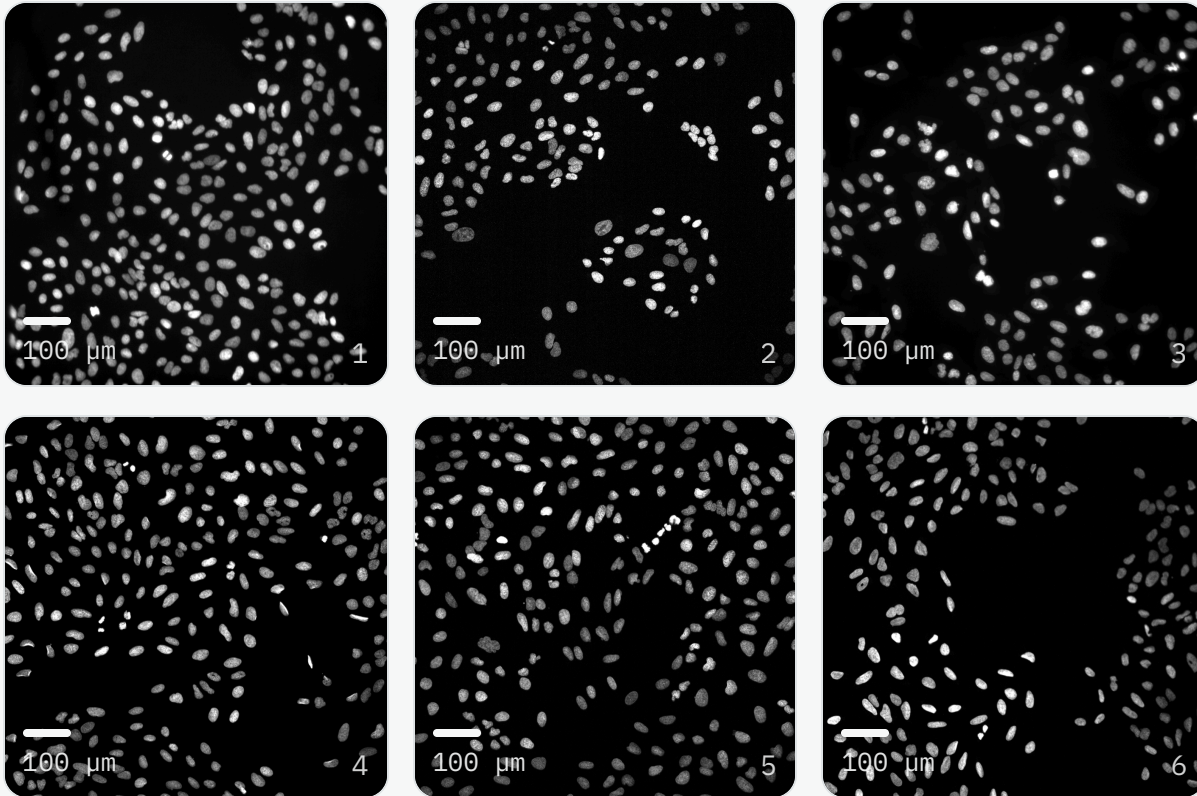
\$28B



**Fig. 1** Checking a new line today: a workflow, not a measurement. Search results are real U2OS images from six papers; sources in the README. [6] Freedman et al., *PLoS Biol*, 2015.

# One lab, or six?

Cell Painting five dyes, 3,773 numbers per well

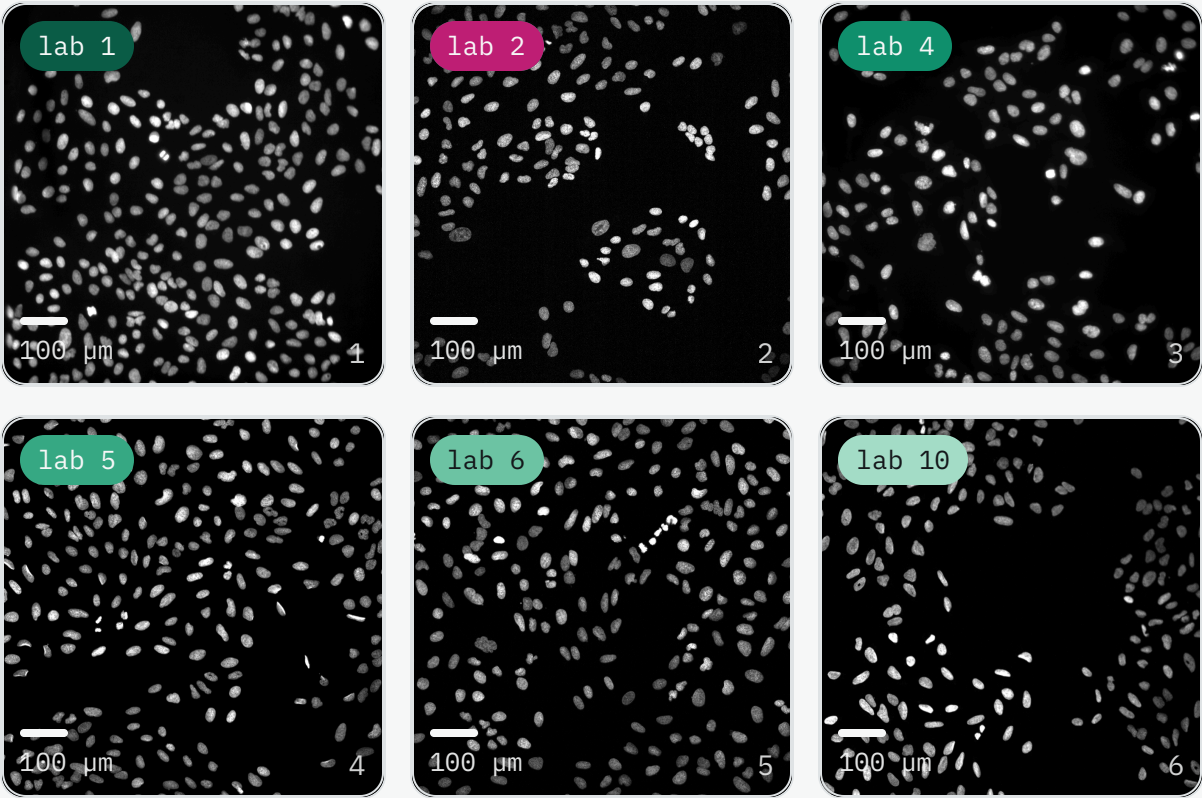


-  same line ✓
-  same protocol ✓
-  no drug ✓

Fig. 3 JUMP cpg0016 (CC0) [7] · 93,228 untreated wells, 11 labs, held out by plate · 99.9% vs chance 9.1% · results/e1

click → the answer

# Could you tell? A classifier can.



Cell Painting five dyes, 3,773 numbers per well

99.9% CLASSIFIER

9% CHANCE

78 wells wrong, of 93,228

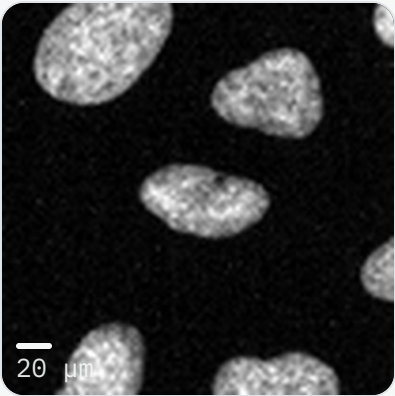
Centre each plate on its own controls: 99.9% becomes 24%. Right for discovery, wrong for QC.

Fig. 3 JUMP cpg0016 (CC0) [7] · 93,228 untreated wells, 11 labs, held out by plate · 99.9% vs chance 9.1% · results/e1

click → follow one well

# One well, **all the way out.**

ONE NUCLEUS



192 μm² · 0.59

ITS WELL

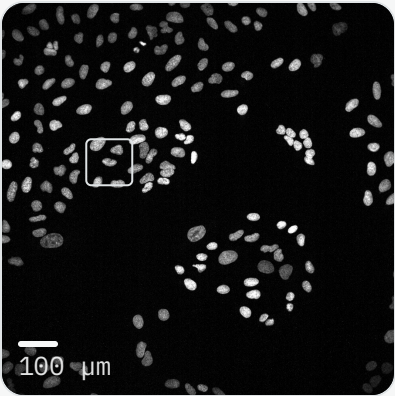
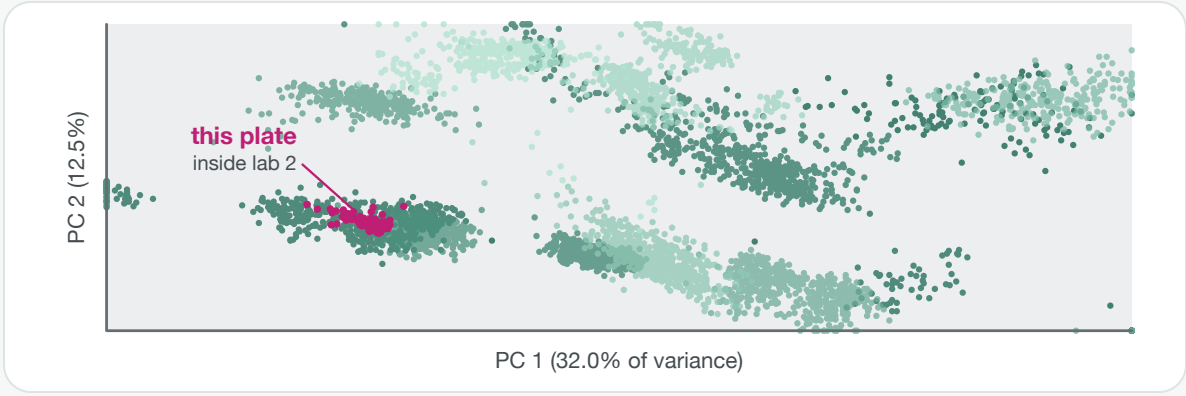


plate 1053600681

3,773  
numbers  
→  
**one point**

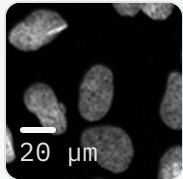
WHERE IT LANDS



93,228 wells, 11 labs

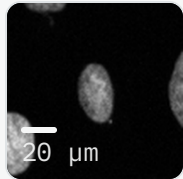
other labs

lab 5



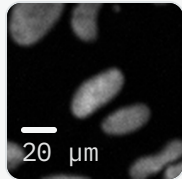
195 μm² · 0.94

lab 6



213 μm² · 0.50

lab 10



207 μm² · 0.29

AREA     ≈ ≈  
TEXTURE    ≠ ≠

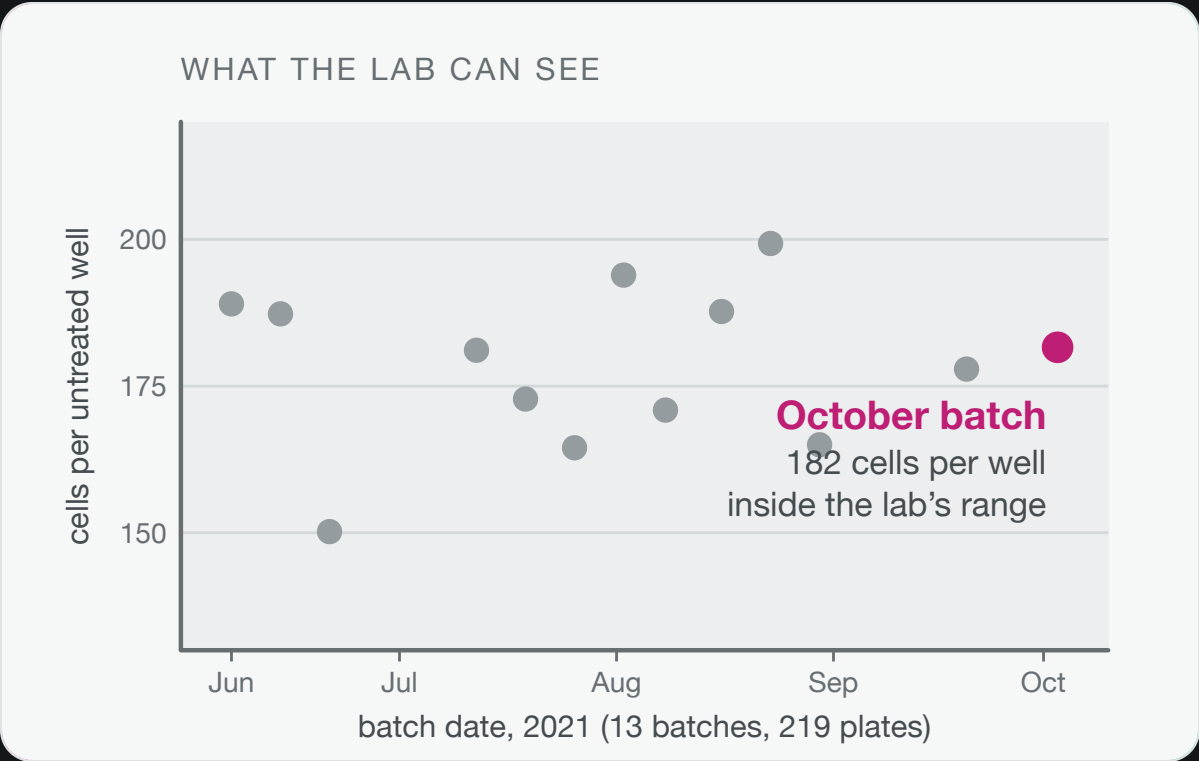
One lab's normal is not another's. The difference is shape, not count.

Fig. 3 JUMP cpg0016 (CC0) [7] · 93,228 untreated wells, 11 labs, held out by plate · 99.9% vs chance 9.1% · results/e1

[click → start over](#)

# It passed every check.

✔ local QC passed



Cells per untreated well, lab 2, 13 batches, June to October 2021. Click for what only the field can see.

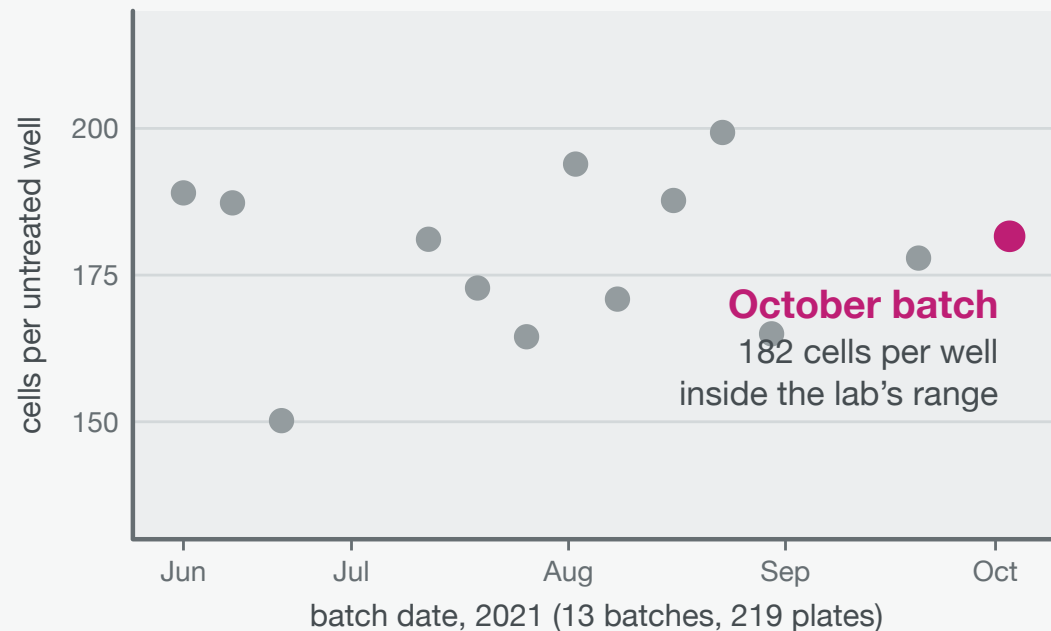
click → the field

# Four months on, nearer lab 8 than itself.

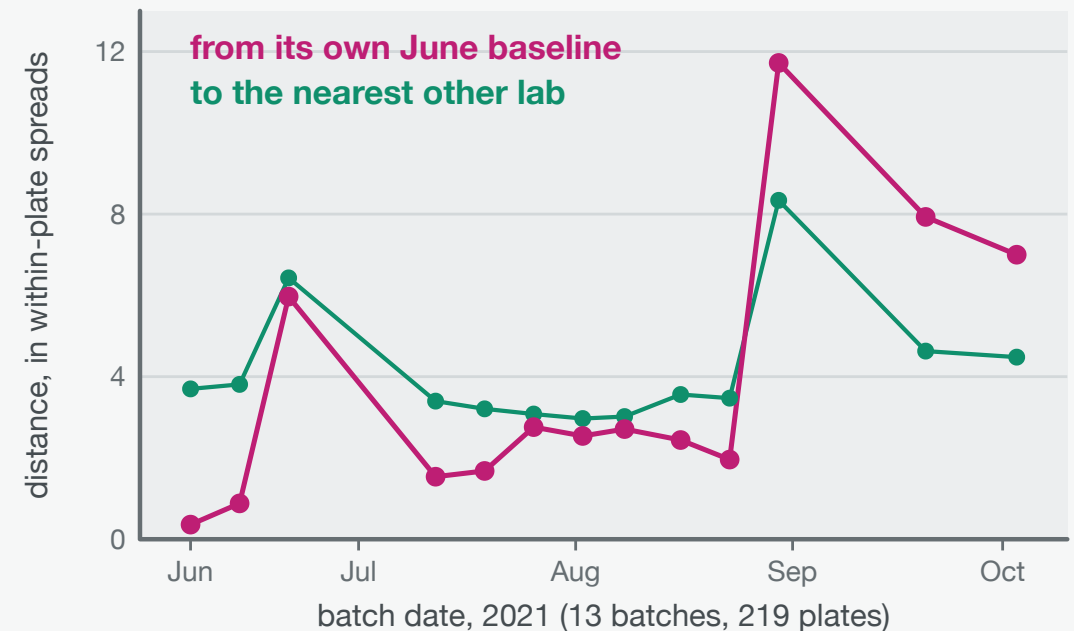
# 7.0

spreads from its own June baseline  
4.5 to lab 8  
still named lab 2 by every well

WHAT THE LAB CAN SEE



WHAT ONLY EVERYONE ELSE CAN SEE



**Fig. 4** Left: mean cells per untreated well, per batch. Right: median distance of each batch's plates from the lab's first batch, and to the nearest other lab's centre, in within-plate spreads; z-scores clipped to  $\pm 5$ . 93,228 wells, 11 labs, 3,773 features [7]. Fluorescence; brightfield is the bet. results/e2-timeline

click → the map

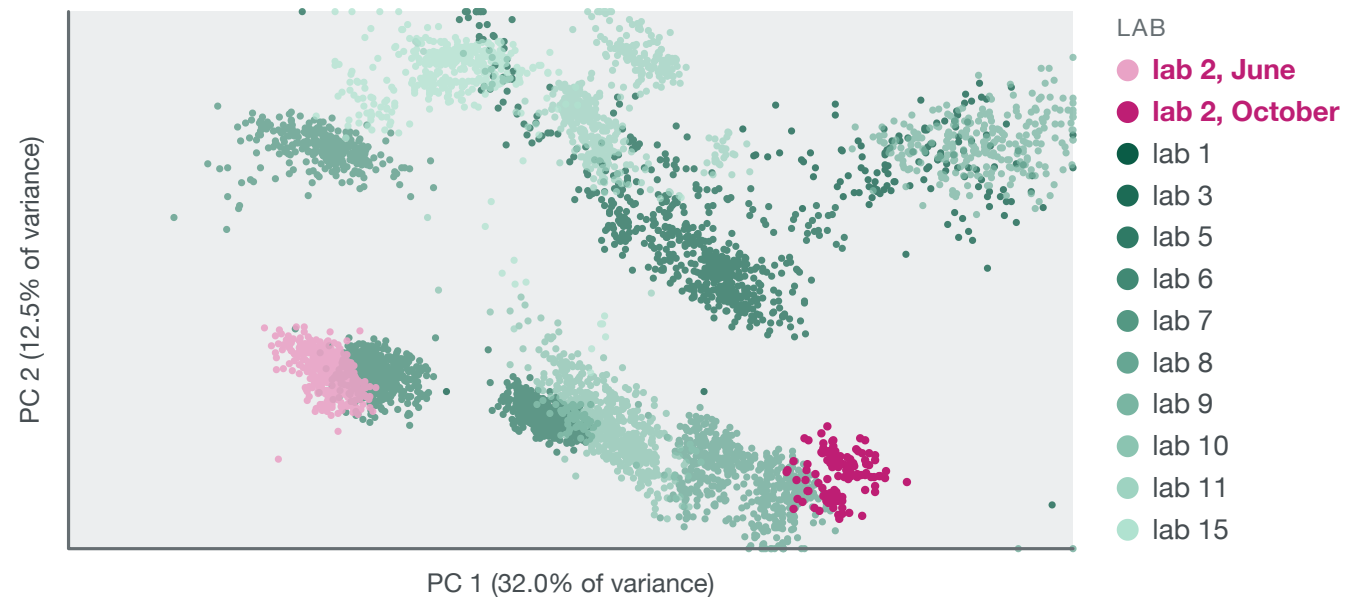
# Same lab, **two different places.**

Same lab, same line, same protocol.

# 7x

two plates of one lab sit  
0.6 spreads apart in the same batch  
4.7 apart over 13 weeks later

EVERY UNTREATED WELL, COLOURED BY LAB



Your own past is not a reference. The field is.

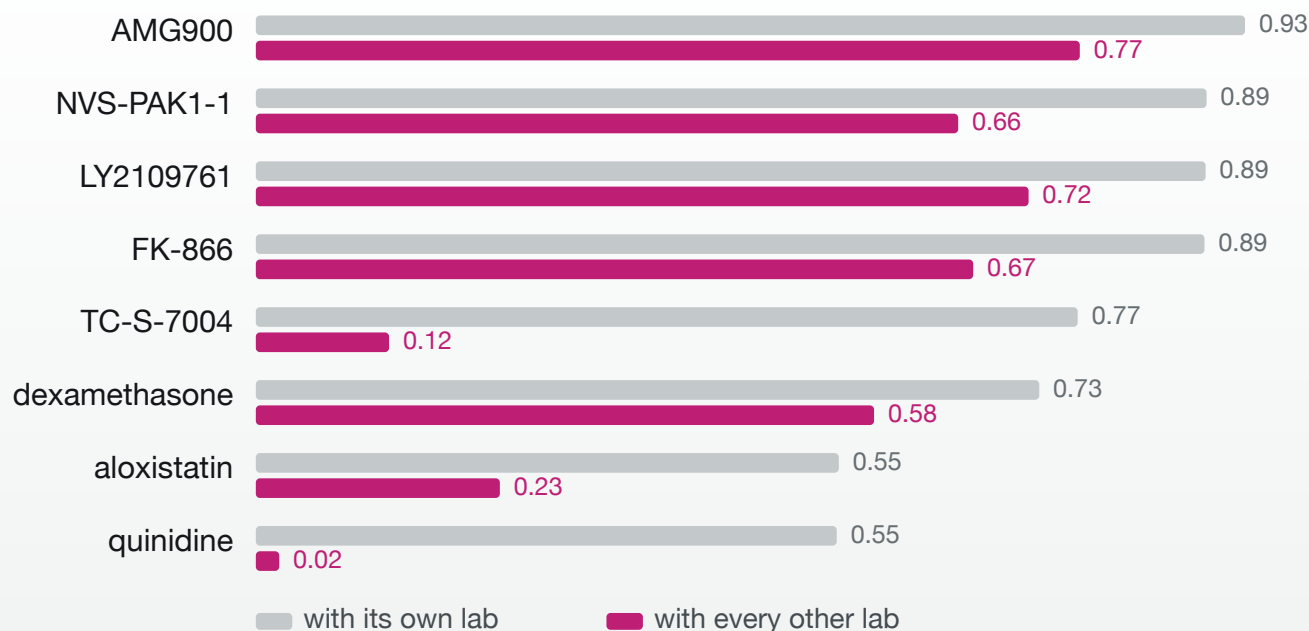
**Fig. 5** PCA of 15,352 untreated wells, 8 per plate. Distances are measured in all 3,773 features, not in this projection.  
results/e2, e1b

click → the drug

# The same drug, a different answer.

## HOW MUCH LABS AGREE ON WHAT A DRUG DOES

agreement of a plate's measured effect (cosine), 0 to 1



## AND WHY

# 16–41%

how much the same compound's effect size differs between labs

inside one lab: 9–32% (6 compounds)

**The further a plate's controls sit from the field, the more its answer moves.**

inside labs: Spearman 0.35, shuffled 0.01

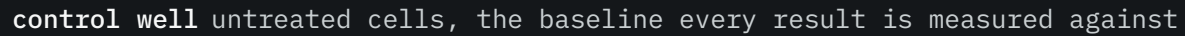
between 11 labs: same sign, too few to prove

Your controls' position predicts how much your answer moves. This is the failure Datum Line was founded on: profiles that would not align to the field's reference.

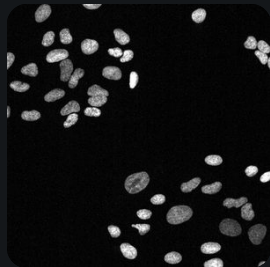
**Fig. 6** JUMP's eight shared positive controls, 275 plates, 11 labs; effect = treated minus same-plate DMSO; each plate compared with every lab but its own [7] · [results/e8-redteam](#)

[click → start over](#)

Send the control-well images you already take. Back comes a certificate, **and which features moved.**



in distribution inside the spread of every other lab



BRIGHTFIELD · THE BET · DNA · THE EVIDENCE

datum line ● 1 day

LAB 2 • OCTOBER BATCH



U2OS, untreated wells, 14 plates, 3  
October 2021

93,228 untreated wells from 11 labs [7]

( in distribution

outside it

7.0 spreads from your June baseline, 4.5 from lab 8. Cell count normal.

```
[e6, pending]
```

Held by nobody who sells you cells,  
media or the instrument.

# Identity, yes. Behaviour, no.

|                                         | STR authentication | Vendor certificate | Plate's own controls | Phenomics platforms | Datum Line |
|-----------------------------------------|--------------------|--------------------|----------------------|---------------------|------------|
| Which line it is                        | ✓                  | ✓                  | —                    | —                   | ✓          |
| How it behaves, the whole cell          | —                  | —                  | ✓                    | ✓                   | ✓          |
| Against every lab running the same line | —                  | —                  | —                    | ✗                   | ✓          |
| From images you already take            | —                  | —                  | ✓                    | —                   | ✓          |
| Which way to move them back             | —                  | —                  | —                    | —                   | ✓          |
| A bad lot, flagged across labs          | —                  | —                  | —                    | —                   | ✓          |

✓ yes    ✓ with a partner: run STR    ✗ removed by design: these platforms correct lab differences away

# Three buyers, one certificate.

The labs that build the reference use it free. The labs that cannot afford to be wrong pay for it.

|                  |  ACADEMIC LABS AND CORES |  PHARMA AND CRO SCREENING |  CELL BANKS AND VENDORS |
|------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| What is at stake | the paper                                                                                                   | the screen · ~\$2.6M                                                                                         | the lot                                                                                                    |
| What they pay    | free, when they contribute control images                                                                   | a subscription per cell line per quarter, ~\$5–15k<br>170× less than one unchecked screen                    | a fee per certified lot                                                                                    |
| What it builds   | the reference                                                                                               | the revenue                                                                                                  | the network: bad-lot alerts, later                                                                         |

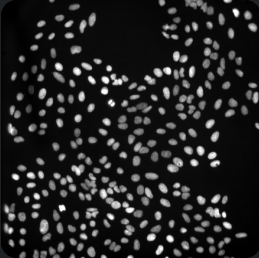
Proposed, not yet validated. Bad-lot alerts across the network come after the certificate. Details in docs/roadmap.

# Build the ugliest dataset in biology, on purpose.

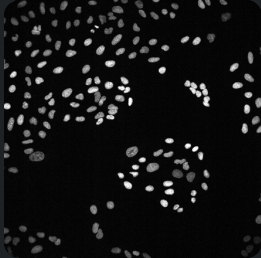
The variation a real lab meets, generated rather than avoided. Same cells, same protocol, no drug, and it already looks like this.

THE TEST SET · WHAT SIX LABS ALREADY SEND

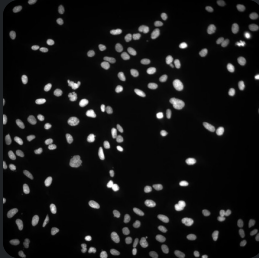
real wells, JUMP cpg0016, CC0



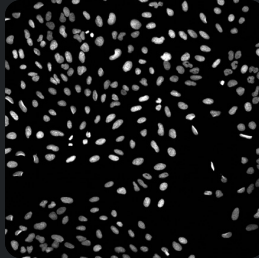
LAB 1  
Phenix · widefield  
NA 1.0 · 0.60 µm



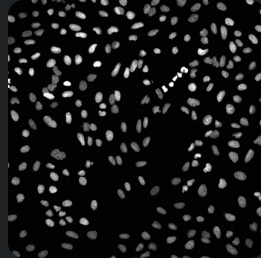
LAB 2  
CV8000 · confocal  
NA 1.0 · 0.65 µm



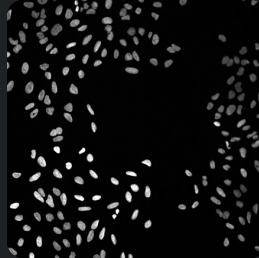
LAB 4  
Phenix · widefield  
NA 1.0 · 0.60 µm



LAB 5  
CV8000 · confocal  
NA 0.75 · 0.65 µm



LAB 6  
CV8000 · confocal  
NA 0.75 · 0.65 µm



LAB 10  
CV8000 · confocal  
NA 0.75 · 0.65 µm

5  
microscope models

0.4–1.0  
objective NA

0–3  
brightfield planes

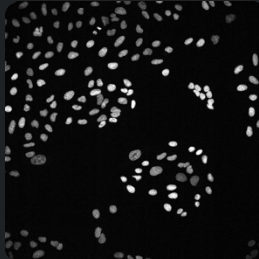
0.60–0.70  
µm per pixel

8  
plate types

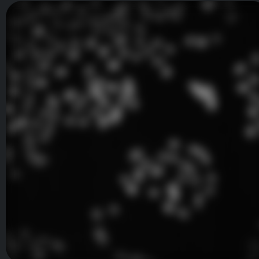
Across the 13 JUMP sites. A real lab adds cell lines, plate types, stacks and damaged plates on top.

RED TEAM

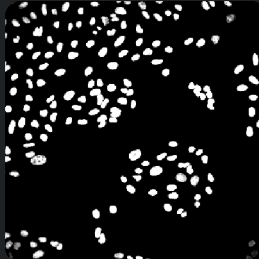
one well, simulated



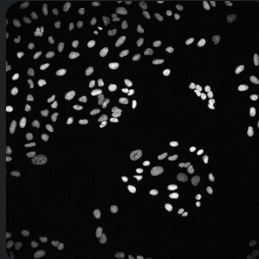
AS IMAGED



DEFOCUSED



OVER-EXPOSED



CLIPPED WELL

A public benchmark: name the failure.  
Bubble, focus, clipped well, bad lot, wrong label.

A foundation other people can stand on: one phenotypic space the field converges toward, instead of Google Images.

# Every plate you image, placed in the field.

A growth chart for cell lines. Every certificate adds a plate to the reference, and the reference is what nobody can build alone.

## WHERE THIS GOES



### NEXT

what to change, which lots work



### THE SET

a reference built to break



### THEN

bad-lot alerts across labs



### BEYOND

microscopes · catalogue · drift

## WHAT I NEED



**Images** U2OS controls



**People** ML, normalization



**Bench** 3-site proof, then the set



**Pre-seed** to run it

---

**Have images of U2OS cells? Talk to me.**

# The rest, if you ask.

Find the question, jump to the board. Every number traces to results/e1, e1b, e2 or a numbered reference; pending work says pending.

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**A1**      **Why not trust the plate’s controls?**  
There is no control well

---

**A2**      **Show me the math.**  
Six operations on one embedding

---

**A3**      **Why not batch-correct it away?**  
What JUMP records

---

**A4**      **Is the model the moat?**  
Counting cells often wins

---

**A5**      **Why not just count cells?**  
Two ways to ask one question

---

**A6**      **How often do I re-certify?**  
The line is not expired

---

**A7**      **Isn’t this STR?**  
STR says who, images say how

---

**A8**      **Is the assay itself working?**  
A known answer

---

**A9**      **Does it change the result?**  
Same drug, different answer

---

**A10**      **Does it work without stains?**  
Bet on brightfield

# There is no control well.

Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole institution. A control only sees inside its own box.

A REAL JUMP PLATE · 384 WELLS

● DMSO 32

● positive 32

○ compounds 320

The pink wells are the plate's zero: 32 of 384, near the edge by design.

A HEALTHY PLATE

A PLATE WHERE EVERYTHING DRIFTED

A control only sees inside its own box.

Whole plate off: its controls move with it.

Whole batch off: so do every plate's.

Whole site off: nobody in the building can tell.

Only the field sits outside every box.

$$x_{test} = \mu + s + \tau(s) \quad x_{ctrl} = \mu + s \quad \Delta = \tau(s)$$

$s$ , the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

**Fig. 2** Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (source\_2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, site and field. No measurements shown.

# Six operations on one embedding.

Every well becomes a vector  $e \in \mathbb{R}^{3773}$ , z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

## 1 · EMBEDDING

**[1]**

$$e = \varphi(\text{image}); \varphi \in \{\text{CellProfiler}, \text{CNN}, \text{ViT}\}$$

Which  $\varphi$  barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

## 2 · LAB CLASSIFIER

**99.9%**

$$f: e \rightarrow \text{lab}, \text{ held out by plate}$$

Logistic regression, L2, five folds. Balanced accuracy, chance 1/11. Permuted labels 9.3%. Without the 1536-well labs: 100.0%.

## 3 · DISTANCE, THE CERTIFICATE

**[e6]**

$$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$$

Difference of two means in units of the field's spread, conditioned on  $c$ . Its direction is what moved;  $-d$  is the way back.

## 4 · VARIANCE DECOMPOSITION

**[e5]**

$$\text{Var}(e) = V_{\text{model}} + V_{\text{site}} + V_{\text{batch}} + V_{\text{plate}} + V_{\text{well}}$$

Nested, so shares sum to one. Microscope model separates from site because models are shared across sites.

## 5 · REGRESS VARIABLES OUT

**24%**

$$e_{\perp} = e - \hat{E}[e \mid c], \text{ then re-run } f$$

Remove only what you know is instrument, keep what may be biology. Remove each plate's own control mean and  $f$  falls to 24%.

## 6 · MORPHOLOGY CHANGE, TWO WAYS

 **$\delta$** 

$$\text{DISTANCE } \delta = f(\|e - E[e \mid \text{vehicle}]\|), f \text{ bounded}$$

$$\text{CLASSIFIER } g: e' \rightarrow \text{label}, e' \text{ aligned to the reference before comparison}$$

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

# What JUMP records. What nobody does.

Normal is a conditional expectation,  $E[x \text{ / every variable below}]$ . Even JUMP, built to be comparable, records only the instrument side.

RECORDED IN JUMP

- site
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- sites per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze-thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

NORMALISE THE LAB AWAY AND THE LAB DISAPPEARS

- 99.9% raw features  
balanced accuracy naming the lab
  - 24% plate-centred  
own control mean removed; chance 9%
  - 19% plate-centred and scaled  
own control spread removed too
- 29,943 wells, 16 per plate, plates held out. results/e1-redteam.

Centre each plate on its own controls, the standard first step, and the lab is named 24% of the time. It removes most of the coordinate a certificate needs.

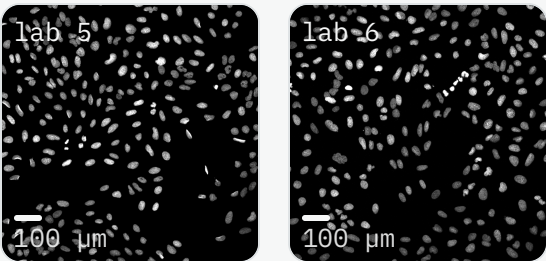
# And the benchmarks are so weak that **counting cells** often wins.

If a naive baseline matches your model, the benchmark is the problem. The bottleneck was never the model. It is what you are comparing against.

# Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



|            |      |      |           |
|------------|------|------|-----------|
| cell count | 236  | 224  | same      |
| shape      | 0.94 | 0.50 | different |

METHOD ONE

Does this run belong with the past runs?

$$g: e' \rightarrow \text{belongs} / \text{does not}, \quad e' = \text{align}(e)$$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

How far out, and which way back?

$$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$$

Bounded, so one wild well cannot swamp the score.

Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else’s images. It takes a reference to know which.

**Fig. 1** Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

# The line is not expired. Your baseline is.

Two plates of one lab sit further apart the longer the gap between them. In lab 2 the distance is still rising after thirteen weeks.



**IS THE CELL LINE RUINED?**  
Not known from profiles alone. The data says the lab moved and when. It does not say why.

**SO WHY RE-CERTIFY?**  
Because your certificate is a comparison made on one day, and the baseline it was made against goes stale within weeks.

**HOW SOLID IS THE NUMBER?**  
One lab, 219 plates, 13 batches, June to October 2021. 0.63 in the same batch, 4.7 over 13 weeks. Other labs are not yet measured.

Median distance between plate centroids, in within-plate spreads, with the interquartile range. results/e1b-drift.

# STR says who.

## Images say how.

Not a competitor: the other half of the same certificate. Every lab already runs the left column. The right column costs nothing extra.

|                | STR PROFILING                                       | CONTROL-WELL IMAGES                                          |
|----------------|-----------------------------------------------------|--------------------------------------------------------------|
| WHAT IT READS  | DNA at 17 short tandem repeat loci                  | the pictures the screen already took                         |
| WHAT IT PROVES | the line is what the label says                     | the cells still behave like the field's version of that line |
| WHAT IT MISSES | two cultures, same fingerprint, different behaviour | identity: it cannot tell you the line is mislabelled         |
| SAMPLE         | cells spotted on a card and mailed                  | nothing leaves the building but images                       |
| EXTRA WORK     | a separate prep, per line, per test                 | none: control wells you already image                        |
| CADENCE        | on receipt, after 10 passages, when in doubt        | every quarter, from the last plate you ran                   |
| ANSWER         | a profile to compare by eye or by database          | one of two words, and a distance                             |

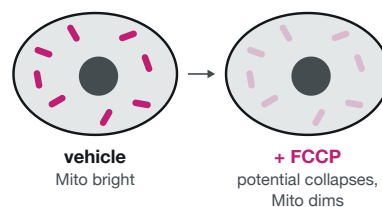
Identity once. Behaviour every plate. One line on the certificate each.

STR practice and cadence: ATCC cell authentication service and test recommendations; 17 loci plus amelogenin on the Promega PowerPlex 18D system. Low- and high-passage LNCaP respond differently to the same hormones: ATCC technical document.

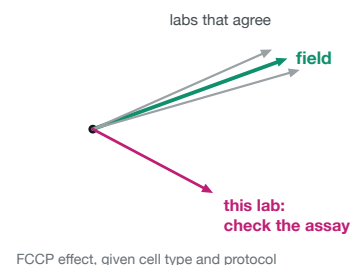
# Next: a known answer, run in every lab.

Controls certify the cells. Positive controls certify the assay. Once cell type and protocol are accounted for, a known perturbation should move cells the same way everywhere.

KNOWN ANSWER



SAME EFFECT, EVERY LAB?



## RUNS NOW ON JUMP

Compound plates already carry 8 positive controls in 32 wells. Same effect at every site?

## ADD ONE: FCCP

MitoTracker loads by membrane potential; FCCP collapses it. Intensity is the obvious readout and the worst one: gain, exposure and filter set differ by lab. The shape change does not.

## THE TEST

$$\delta = E[x \mid pos, c] - E[x \mid vehicle, c]$$

Pass if a lab's  $\delta$  points where the field's does, at a size inside the field's spread. Not “did the channel dim” but “did the cells change the way the field says they change”. Vehicle fails: fix the cells. Positive control fails: fix the assay.

FCCP or CCCP as the loss-of-potential control: Thermo Fisher MitoTracker guidance. MitoTracker Deep Red dims on depolarisation: PMC4811937. Intensity is not comparable across instruments without calibration; morphology is scale-free. Response differs by cell type: Pendergrass et al., Cytometry A, 2004. Plate layout: JUMP metadata.

# Same drug. Different answer.

Every JUMP compound plate runs the same eight positive controls beside its DMSO. So the field already contains the experiment: one fixed compound, measured 1,829 times, in eleven buildings.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{ compound } c] - E[x \mid \text{plate } p, \text{ DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.

Q1 · DOES IT AGREE?

[e8]

Cosine between each plate's  $\delta$  and the field's, within a lab against across labs.

Every compound agrees less with other labs: own lab 0.55 to 0.93, other labs 0.02 to 0.77. AMG900 0.93 against 0.77.

Q2 · DOES IT TRACK THE CONTROLS?

[e8]

Spearman between a plate's control distance and how far its  $\delta$  sits from the field's.

Inside labs, yes: Spearman 0.30 to 0.58 for 6 compounds; pooled 0.35, shuffled 0.01. Between 11 labs: same sign, not established.

Q3 · DOES IT MATTER?

[e8]

Across-lab spread of  $\|\delta\|$  as a percentage of the effect itself.

Effect size differs 16% to 41% between labs, 9% to 32% inside one. quinidine and TC-S-7004 are too weak to measure.

Q2 is positive inside every lab: your answers differ. 275 plates, 25 per lab, 11 labs; each plate against every lab but its own.

1,829 compound plates carrying both poscon and DMSO, 11 sources, 129 batches; every compound on at least 1,730. 144,279 wells. Positive controls: JUMP perturbation\_control.csv. Scripts: analysis/e8\_pull\_wells.py, analysis/e8\_same\_drug\_different\_answer.py, analysis/e8\_redteam.py.

# Shown in fluorescence. Bet on brightfield.

Every number here is Cell Painting: five stains. The product reads brightfield: none. That gap is the bet, testable on public data.

## WHAT IS SHOWN

### Stained wells name the lab

Untreated U2OS wells, 11 labs: the lab is named 99.9% of the time, chance 9%.  
results/e1

## WHY THE BET IS FAIR

### Fewer variables, not more

No stain lot, dye lot or staining step:  
three unrecorded variables gone (A3).  
JUMP images brightfield beside the  
stains, so the same wells re-run without  
them.

## WHAT WOULD KILL IT

### Brightfield alone at chance

Then the certificate asks for one cheap  
stain. The test: e1 on the brightfield  
planes of the same 93,228 wells.  
[pending]

Brightfield timecourse is the modality I build screens on.